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Evaluation of NASA OPERA DIST-ALERT Data for Monitoring Vegetation Disturbance in Miombo Forests, Zambia

*Case Study: Mabonga Community Forest Area (CFA),
Lavushimanda Community Conservation Project, Zambia*

Research Project Report

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1. Lavushimanda Community Conservation (LCC)

- The Lavushimanda Community Conservation (LCC) project in Zambia's Miombo Belt is crucial for biodiversity, carbon storage, and water regulation, as it links two large national parks (the Lavushimanda NP and the South Luangwa NP).
- Foreseen restoration and sustainable management of ca. 30,000 hectares of Miombo woodlands across a combination of community forest areas (CFAs) and private lands, aiming to enhance biodiversity, carbon sequestration, and local livelihoods. Approach:
 - Community Engagement: Collaborate with local communities to involve them in sustainable forest management and conservation of the CFAs and forest land on farms.
 - Conservation and assisted natural regeneration of (degraded) Miombo forest through fire management: Employ fire prevention and control strategies to manage the land under restoration, addressing the issue of uncontrolled fires that might hinder forest recovery.

2. Purpose of this Analysis

The NASA OPERA Surface Disturbance (DIST-Alert) data product is a near-global satellite-based system designed to detect land surface disturbances at high spatial and temporal resolution, using integrated optical and radar imagery from multiple satellite missions. It provides disturbance information related to processes such as fires, deforestation, drought, urban expansion, and other land cover changes. The differenced Normalized Burn Ratio (dNBR) index is a widely established remote sensing indicator used to quantify burn severity and assess fire impacts on vegetation (Key and Benson, 2006; USGS, 2018).

The purpose of this analysis is to evaluate the suitability and effectiveness of NASA OPERA DIST-Alert data for detecting and quantifying fire-induced forest disturbances in the Mabonga Community Forest Area (CFA), Miombo forests in Zambia, which constitute a key area within the Lavushimanda Community Conservation (LCC) project.

The specific objectives of this analysis are:

1. To assess the capacity of DIST-Alert data to identify and quantify fire-related forest disturbances by comparing its outputs with burn severity information derived from the dNBR index.
2. To examine the potential of near-real-time DIST-Alerts as an operational tool for forest monitoring and management within the LCC project, supporting the prevention of illegal land degradation and large-scale forest fires.

3. To evaluate the consistency between DIST-Alert detections and dNBR-derived burn severity patterns, to determine the reliability of DIST-Alert data for monitoring fire-induced disturbances in Miombo forest ecosystems.

3. NASA OPERA Surface Disturbance (DIST-Alert) Data

NASA Observational products for End-Users from Remote Sensing Analysis (OPERA) consists of four focus areas: Surface Water Extent, Surface Disturbance, Surface Displacement, and Vertical Land Motion¹. The Surface Disturbance (DIST) Product is a Level 3 product that detects changes in the Earth's surface using harmonized and Landsat 8 and Sentinel-2 A/B (HLS) (optical data). DIST-ALERT maps per-pixel vegetation disturbance, specifically, vegetation-cover loss. Vegetation disturbance is mapped when there is an indicated decrease in vegetation cover within an HLS pixel. In addition, DIST-ALERT provides auxiliary generic disturbance² information derived from variations in reflectance across the HLS³ scenes, capturing general disturbance trends. This dataset is publicly available through NASA's Land Processes Distributed Active Archive Center (LP DAAC) (Pickens et al., 2024).

3. 1 Dataset Description

The OPERA Project generates DIST-ALERT using validated Level-1 data products from existing missions (NISAR, SWOT, Sentinel-1) and projects (HLS). It does not generate Level-0 or Level-1 products directly. The dataset is sponsored by NASA and developed by the Jet Propulsion Laboratory (JPL). DIST-ALERT provides near-global coverage, including per-pixel land surface change from HLS scenes, at 30 m spatial resolution with a revisit period of approximately 2–3 days. The dataset first became operational in April 2021 and is distributed in GeoTIFF file format. DIST-ALERT comes in two variants according to temporal scope (Pickens et al., 2024)

- DIST-ALERT: generated at the cadence of HLS imagery, providing near-real-time alerts.
- DIST-ANN: a summary of DIST-ALERT over the previous year, showing confirmed changes.

3.2 Bands Overview

DIST-ALERT contains 19 bands. Table 1 is a detailed description based on public documentation and the dataset specification:

¹ OPERA Focus Areas: The four domains monitored by NASA OPERA for land surface change.

² Auxiliary Generic Disturbance: A detected land surface change that indicates disturbance has occurred, but cannot yet be attributed to a specific cause based on available spectral or temporal data.

³ HLS: Harmonized Landsat Sentinel-2.

Raster Layer Description	Layer Name	Data Type	Layer Values
Vegetation Disturbance Status Indication of vegetation cover loss (vegetation disturbance). Status labels based on maximum anomaly value, confidence, and whether ongoing or finished: "First", "Provisional", "Confirmed", "Finished". Applies to both above/below 50% disturbance threshold.	VEG-DISTSTATUS	UInt8	0: No disturbance 1: First <50% 2: Provisional <50% 3: Confirmed <50% 4: First ≥50% 5: Provisional ≥50% 6: Confirmed ≥50% 7: Confirmed <50%, finished 8: Confirmed ≥50%, finished 255: No data
Current Vegetation Cover Indicator Percent vegetation cover estimated for current HLS scene.	VEG-IND	UInt8	0–100: Estimated percent

			vegetation 255: No data
Current Vegetation Anomaly Value Difference between historical baseline and observed vegetation cover (0–100% loss). Sum with current cover gives historical vegetation estimate.	VEG-ANOM	UInt8	0–100: Estimated vegetation loss 255: No data
Historical Vegetation Cover Indicator Historical percent baseline at time of maximum anomaly; fill for non-disturbance pixels. Calculated from HLS scenes within ± 15 days over previous three years.	VEG-HIST	UInt8	0–100: Vegetation percent 200: No disturbance 255: No data
Max Vegetation Anomaly Value Maximum decrease in vegetation cover; can threshold disturbances (e.g., $\geq 20\%$).	VEGANOMMAX	UInt8	0–100: Maximum loss of vegetation percent 255: No data
Vegetation Disturbance Confidence Layer Mean anomaly $\times$ number of loss anomalies squared. Confidence updated until anniversary or consecutive non- anomalies.	VEG-DISTCONF	Int16	-1: No data 0: No disturbance >0: Disturbance confidence
Date of Initial Vegetation Disturbance Day of first loss anomaly detection, days since Dec 31, 2020.	VEG-DISTDATE	Int16	-1: No data 0: No vegetation anomalies in last year

			>0: Day of initial anomaly detection
Number of Detected Vegetation Loss Anomalies Total anomalous observations since first detection (max 254).	VEG-DISTCOUNT	UInt8	0: No disturbance 1–254: Count of loss anomalies 255: No data
Vegetation Disturbance Duration Days of ongoing loss anomalies (max one year).	VEG-DISTDUR	Int16	-1: No data 0: No disturbance 1–366: Days from first to most recent anomaly
Date of Last Observation Assessed Last quality-assessed HLS observation used for baseline calculation.	VEG-LASTDATE	Int16	-1: No data ≥1: Last day assessed
Generic Disturbance Status* Indication of generic spectral difference. Labels: First, Provisional, Confirmed, Finished. Applies to low/high spectral anomaly thresholds.	GEN-DISTSTATUS	UInt8	0: No disturbance 1: First low 2: Provisional low 3: Confirmed low 4: First high 5: Provisional

			high 6: Confirmed high 7: Confirmed low, finished 8: Confirmed high, finished 255: No data
Current Generic Disturbance Anomaly Value Spectral distance from current HLS scene to previous three-year baseline (Mahalanobis distance).	GEN-ANOM	Int16	-1: No data 0: No disturbance >0: Spectral distance
Generic Disturbance Maximum Anomaly Value Maximum spectral distance relative to historical composite.	GENANOMMAX	Int16	-1: No data 0: No disturbance >0: Spectral distance
Generic Disturbance Confidence Layer Mean spectral distance × number of anomalies above threshold; updated until anniversary or consecutive non-anomalies.	GEN-DISTCONF	Int16	-1: No data 0: No disturbance >0: Disturbance confidence
Date of Initial Generic Disturbance Day of first spectral anomaly detection, days since Dec 31, 2020.	GEN-DISTDATE	Int16	-1: No data 0: No spectral anomalies

			in last year >0: Day of initial anomaly detection
Number of Detected Spectral Anomalies Total spectral anomaly observations (max 254).	GEN-DISTCOUNT	UInt8	0: No disturbance 1–254: Count of loss anomalies 255: No data
Generic Disturbance Duration Days of ongoing spectral anomalies (max one year).	GEN-DISTDUR	Int16	-1: No data 0: No disturbance 1–366: Days from first to most recent anomaly
Date of Last Observation Assessed for Generic Disturbance Last quality-assessed HLS observation used for baseline calculation.	GEN-LASTDATE	Int16	-1: No data ≥1: Last day assessed
Data Mask* Mask of pixels where algorithms are applied; based on Fmask of HLS granule.	DATAMASK	UInt8	0: Not land 1: Land 2: Water

Table 1 : DISTALERT Raster Layers for v1 (source: Pickens et al., 2024)

The bands used to monitor forest loss or vegetation disturbance alerts are Date of Initial Vegetation Disturbance (VEG-DISTDATE), Number of Detected Vegetation Loss Anomalies (VEG-DISTCOUNT), and Maximum Vegetation Anomaly Value (VEG-ANOMMAX).

3.3 Global Forest Watch (GFW) Approach

GFW integrates DIST-ALERT for global disturbance monitoring (Figure 1). Each pixel (alert) encodes the date of disturbance and confidence level in one integer value. The leading integer of the decimal representation is 2 for a low-confidence alert (2-3 consecutive observations), or 3 for a high-confidence alert (4 or more consecutive observations), followed by the number of days since December 31, 2020. 0 is the no-data value. For example:

- 20001 is a low confidence alert on January 1st, 2021
- 30055 is a high confidence alert on February 24, 2021
- 31701 is a high confidence alert on August 28, 2025
- 0 represents no alert

The layers: VEG-DISTDATE to filter the data by the date, and to determine whether to update pixels when there are multiple alerts in the same location; VEG-DISTCOUNT is used to determine the confidence (2-3 consecutive observations for low confidence (represented by number 2) and 4+ observations for high confidence (represented by number 3); and VEG-ANOM-MAX is used as a threshold for determining loss (>30%).

In this study, the DIST-Alert dataset was accessed through the Global Forest Watch (GFW) platform and processed in Google Earth Engine (GEE). Alerts were filtered spatially using the Mabonga Community Forest Area (CFA) boundary shapefile and temporally for the study period between July and October 2025. The filtered alerts were aggregated at bi-weekly intervals, and the spatial extent of disturbance was calculated for each monitoring period. The total disturbed area was expressed in hectares and as a percentage of the total Mabonga Community Forest Area (CFA) Area (8,199.71 ha) to evaluate the temporal progression of forest disturbance.

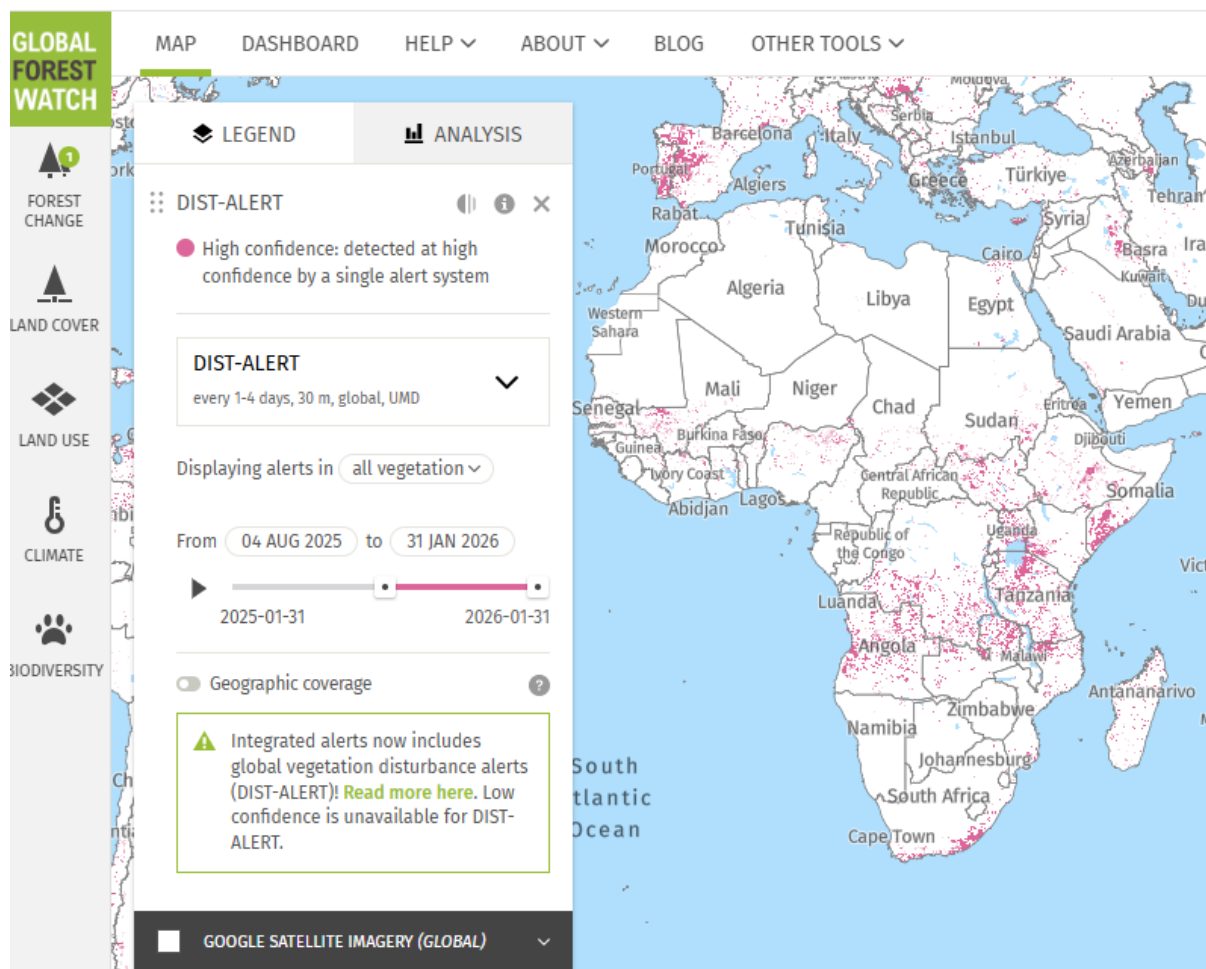


Figure 1: Global Forest Watch DIST-Alert dataset overview

Figure 1 illustrates the structure of the Global Forest Watch DIST-Alert dataset used in this study. The dataset integrates NASA OPERA surface disturbance alerts and encodes both the date of disturbance and confidence level within a single pixel value, allowing for near-real-time monitoring of vegetation disturbance events.

4. Materials and Methods

4.1 Study Area

The study was conducted in Mabonga Community Forest Area (CFA), located in Lavushimanda District, in Zambia. Mabonga Community Forest Area (CFA) lies within the northeastern part of the country, in Muchinga Province, which is characterized by extensive miombo woodlands, savanna grasslands, and relatively low population density.

The area experiences a tropical savanna climate, with a distinct rainy season from November to April and a dry season from May to October. Annual rainfall is moderate, supporting both subsistence and small-scale commercial agriculture. Temperatures are generally warm throughout the year, with cooler conditions during the dry season.

Mabonga Community Forest Area (CFA) and its surrounding communities rely mainly on rain-fed agriculture, forestry-related activities, and small-scale livestock keeping for their livelihoods. Major crops grown in the area include maize, groundnuts, beans, and cassava. The natural vegetation is dominated by miombo woodland species, which play an important role in providing fuelwood, construction materials, and non-timber forest products.

The district is relatively remote, with limited infrastructure and access to major markets. Road networks are mostly unpaved, and access to social services remains a challenge in some parts of the area. Despite these constraints, the region possesses significant ecological value due to its forest cover and biodiversity.

The selection of Mabonga Community Forest Area (CFA) as the study area was based on its ecological importance, land-use characteristics, and relevance to the objectives of this research. The area provides a suitable setting for examining environmental and socio-economic interactions, particularly in relation to land management, resource use, and conservation practices.

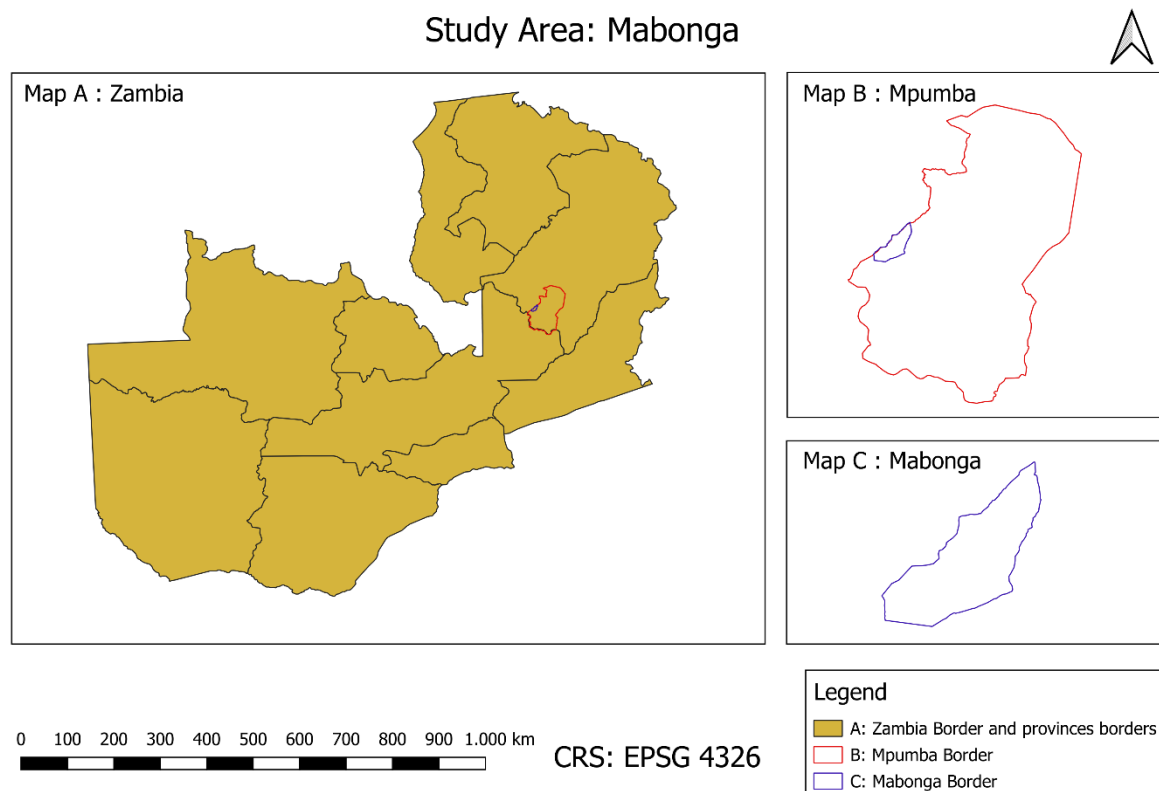


Figure 2 : Mabonga Community Forest Area (CFA) Border Study Area

Figure 2 shows the spatial extent of the Mabonga Community Forest Area (CFA) used as the area of interest for clipping DIST-Alert and Sentinel-2 datasets. This boundary was applied to extract disturbance alerts and calculate the affected area within the study region.

4.2 Data

4.2.1 DIST-Alert Data

The DIST-Alert bands used in this study are the Date of Initial Vegetation Disturbance(VEG-DISTDATE) and Number of Detected Vegetation Loss Anomalies(VEG-DISTCOUNT). Date of Initial Vegetation Disturbance(VEG-DISTDATE) contains the date for alerts specific to a single date and counts date as a numerical value starting 0 from 31st Dec 2020. Number of Detected Vegetation Loss Anomalies(VEG-DISTCOUNT) contains the number of alerts specific to each date, and count 1 means 1 alert having low confidence, 2 means 2-3 alerts, and 3 means alerts above 4.

For our Analysis, we have restricted our study period from July 2025 to October 2025, however, at the start of the research our main focus was to conduct analysis on 2024 data, but due to the unavailability of DIST Alerts for the duration before 2025, it was mainly not possible to get Alerts for previous years and later compare them to the alerts in QGIS Analysis. However, the research has been conducted in a way that with the availability of data the same methodology can be used on previous years e.g. 2024 and vice versa.

The alerts are calculated for every 2 weeks i.e. from 1st of July to 15th, and from 16th of July to the end of July and same repeats for every month until the end of study period i.e. October 2025.

4.2.2 Sentinel-2

Data from the satellite Sentinel-2 were used for classification and calculating the dNBR index (see further). Sentinel-2 acquires multispectral optical data with a spatial resolution between 10 and 60 m as a function of the band considered. The temporal resolution is 5 days with two twin satellites, S2A and S2B. The multispectral optical data were obtained and processed in Google Earth Engine (GEE), referring to the COPERNICUS/S2_SR_HARMONIZED collection. Sentinel-2 L2 data contains 12 spectral bands. In this work, the bands B8 (842 nm) and B12 (2185 nm), with 10 and 20 m spatial resolution, respectively, were used.

In GEE, clouds were removed using a cloud mask. This mask leveraged the Sentinel-2 Scene Classification Layer (SCL) to identify and exclude non-relevant pixels, such as clouds and shadows. This selective masking ensured that the dataset was free from atmospheric interference, improving the reliability of the results.

4.3 Methodology

4.3.1 DIST-Alert

QGIS Analysis

Using the software QGIS, initially three bands: Vegetation Disturbance Status(VEG-DISTSTATUS) , Number of Detected Vegetation Loss Anomalies (VEG-DISTCOUNT) and Date of Initial Vegetation Disturbance(VEG-DISTDATE) are downloaded for each date with a 2 to 3 days gaps for each time period e.g. if the alerts are being calculated from 1st of July to 15th of July, 2025 then the 3 bands available during for every date from 1st of July to 15th of July, 2025 are downloaded. Special thresholds in the Raster Calculator are set for each band to produce results. VEGETATION DISTURBANCE STATUS(VEG-DISTSTATUS), where 1 is used as a threshold to separate no-disturbance than disturbance pixels. The following equation is used to derive the new raster.

$$\text{'VEG-DISTSTATUS} = 1\text{'}$$

Number of Detected Vegetation Loss Anomalies band is used to count the number of alerts:

- 1 alert: is assigned as 1, and means low confidence,
- 2-3 alerts: are assigned as 2 , and means medium confidence
- > 3 alerts: is assign as 3, and means high confidence.

The below equation is used to derive the results for VEGETATION DISTURBANCE COUNT CONFIDENCE RASTER

$$\text{'Number of Detected Vegetation Loss Anomalies =2 \& Number of Detected Vegetation Loss Anomalies band =3)*2 + (Number of Detected Vegetation Loss Anomalies band >=4)*3'}$$

VEGETATION DISTURBANCE DATE is extracted for the period alerts are being calculated i.e. below equation is used to calculate the specific time from the original raster e.g. from July 1st to July 15th.

$$\text{'VEG-DISTDATE}>= 1643 \& \text{VEG-DISTDATE}<= 1657\text{'}$$

Then all the new generated raster files are multiplied to obtain the final raster.

$$\text{'VEGETATION DISTURBANCE STATUS* VEGETATION DISTURBANCE DATE* VEGETATION DISTURBANCE COUNT CONFIDENCE RASTER'}$$

Global Forest Watch

In Global Forest Watch Dashboard the user first selects forest change menu, and inside that Global Integrated Disturbance Alerts (Figure 3), from that the user can select DIST-Alerts. The user can add a shapefile to search for area of interest or can also zoom in and get the results. In the legend window, the user can choose the time period he wants to get the alerts for and get the results.

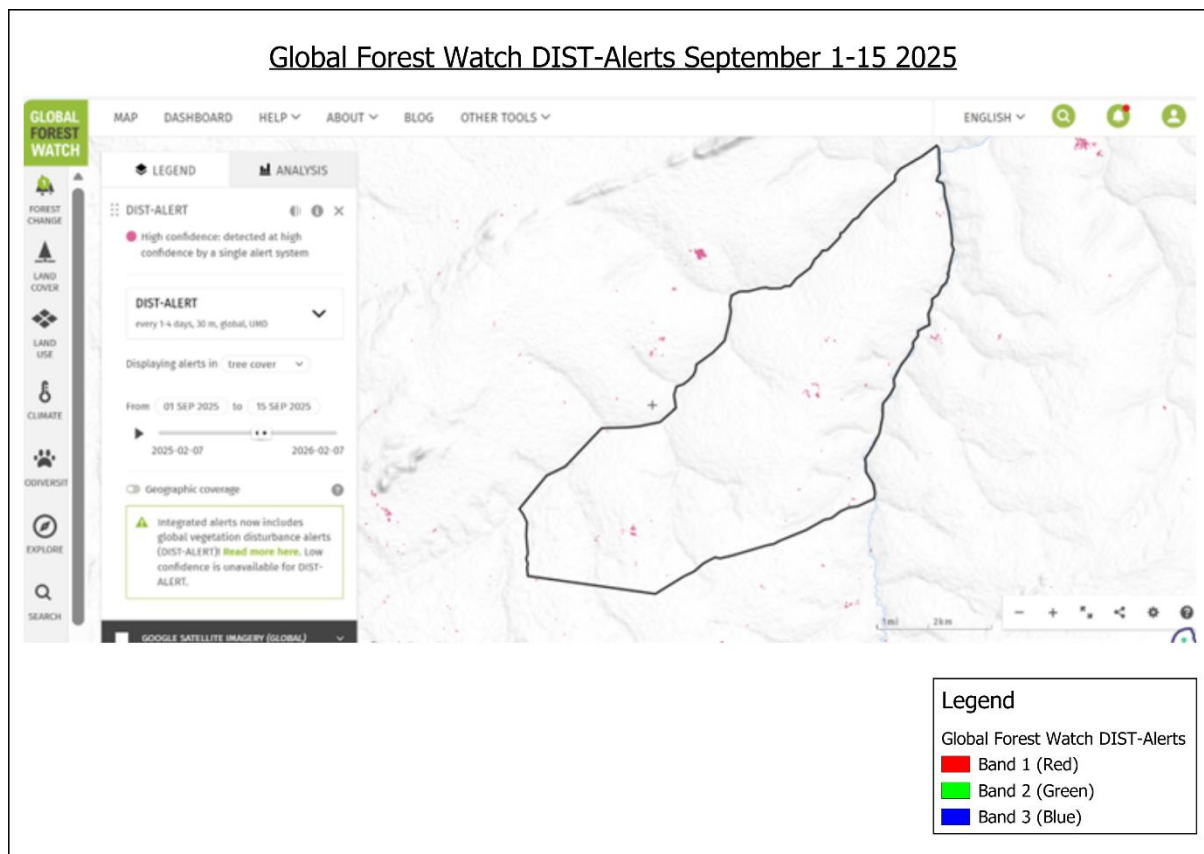


Figure 3 : Figure displaying how to search for DIST-ALERTS in Global Forest Watch for specific time period

Google Earth Engine

The Global Forest Watch is responsible to upload the NASA OPERA DIST-Alert on Google Earth Engine (GEE). On the GEE, the user needs to use 2 bands to derive HLS Disturbance Alerts: Date of Initial Vegetation Disturbance and Number of Detected Vegetation Loss Anomalies. Unlike the Date of Initial Vegetation Disturbance band raster which shows date as a count and in QGIS we have to specifically calculate the digit for each date, in GEE, dates can be specifically entered and filtered.

In Google Earth Engine, first we upload a shapefile in the script then we specify which specific bands we are accessing. In the next step we clip those bands according to our shapefile. Later we specify the base year. For DIST-Alert dataset, we indicate the base year from 31st of December 2020, and it can vary for other datasets, however, for DIST-Alerts it remains same for the entire analysis, then we enter the start date and end date for the DIST- Alerts, and it would only display the alerts during that duration. Later, we count the values for confidence class 1, 2 or 3 for Alerts to display for low, high and highest confidence. We also calculated the area of DIST-Alerts and converted it from hectares to square meters, specified a single projection for the displayed layers.

In the next step, we added the dNBR raster of that period and overlapped the raster with the DIST-Alert raster to find where dNBR values coincide with the DIST-Alert values to do further analysis. We exported 4 separate files of one csv containing area as well as class of alert, another csv file containing the latitude, longitude, dNBR value and the class of alert, same file with kml extension to visualize the values in Google Earth pro and the last DIST-Alert raster for the alerts during that period.

4.3.2 Differenced Normalized Burn Ratio (dNBR) index

The differenced Normalized Burn Ratio (dNBR) index is derived from the Normalized Burned Ratio (NBR) index. The NBR index detects burned areas and assesses vegetation health, especially post-fire. It uses a ratio similar to the Normalized Difference Vegetation Index (NDVI) but integrates the short-wave infrared band (SWIR) instead of the Red band. The SWIRs are strongly sensitive to burned areas, while the near-infrared band (NIR) reflects healthy vegetation (Key and Benson, 2006) (Equation 1).

$$\text{NBR} = (\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR}) \quad (1)$$

NIR is the near-infrared band (band 8), and SWIR is the short-wave infrared (band 12). High NBR values reflect areas covered with healthy vegetation; low values indicate bare ground and recently burned areas, and near-zero values represent areas that are not affected by the fire event.

The dNBR assesses fire severity by quantifying changes in the landscape before (pre-fire) and after a fire (post-fire).

The pre-fire NBR usually represents healthy and unburned vegetation, while the post-fire NBR shows the impacts of fires on the landscape. The dNBR is calculated as Equation 2 below:

$$\text{dNBR} = \text{NBR (pre-fire)} - \text{NBR (post-fire)} \quad (2)$$

To address the lack of free cloud Sentinel-2 images in the specified region of interest, the NBR index was calculated on a bi-weekly basis. This approach allowed for the generation of a consistent dataset. Each month was divided into two periods: the first two weeks (pseudo pre-fire period) and the last two weeks (pseudo post-fire period). By taking the difference between the pre-fire and post-fire NBR values, two distinct types of dNBR were calculated:

- Intra-Month dNBR: This index measures change within the same month, using the NBR difference between the first and last two weeks.
- Inter-Month dNBR: This index captures change across consecutive months, using the NBR difference between the last two weeks of month N and the first two weeks of month N+1.

The United States Geological Survey (USGS) Fire Effects Monitoring and Inventory Protocol (FireMON) program (Lutes et al., 2006) proposed a classification table to interpret the burn severity (based on the dNBR index) for forest fires in North America, which can be seen below (Table 1).

Severity Level	dNBR Range (scaled by 10^3)	dNBR Range (not scaled)
Enhanced Regrowth, high (post-fire)	-500 to -251	-0.500 to -0.251
Enhanced Regrowth, low (post-fire)	-250 to -101	-0.250 to -0.101
Unburned	-100 to +99	-0.100 to +0.99
Low Severity	+100 to +269	+0.100 to +0.269
Moderate-low Severity	+270 to +439	+0.270 to +0.439
Moderate-high Severity	+440 to +659	+0.440 to +0.659
High Severity	+660 to +1300	+0.660 to +1.300

Table 2: Burn severity levels obtained by calculating dNBR, as proposed by USGS.

In this report, the period of Sentinel-2 images used in this work was from July 1 to October 31, 2025.

4.4 Dist-Alert vs. dNBR

The comparison was performed in the Google Earth Engine script using overlap between DIST-Alert and dNBR raster values. During the overlap, the confidence class for the DIST-Alert and the dNBR values at the same location were extracted, to determine whether they quantified for proper alert values and to understand the type of disturbance. A KML file was generated containing locations of the DIST-Alerts, dNBR values as well as the confidence of the alert is displayed. For every 15 days, from July to October 2025, dNBR values were extracted for pixels where DIST-Alert was ≥ 1 .

The above process was repeated for every 15 days, from July 2025 to October 2025, to detect fires, slight forest disturbances during specific periods, forest degradation and conversions at large levels, vegetation recovery as well as checking the confidence level to confirm the disturbance.

Statistics of median, maximum, minimum for dNBR raster values were calculated for Dist-Alert pixels grouped by Confidence classes: 1, 2, and 3 to check the trend for the values.

In the statistics derived, area for DIST-Alerts was also computed in hectares and converted to square metres to check the level of disturbance and compare the results across QGIS and Google Earth Engine.

4.4.1 DIST-Alert Spatial and Temporal Analysis

A temporal analysis of DIST-Alert data was conducted at bi-weekly intervals between July and October 2025 in order to evaluate disturbance progression within the Mabonga Community Forest Area (CFA). The spatial extent of disturbance alerts was calculated for each monitoring period and expressed as a percentage of the total CFA area (8,199.71 ha). These results were subsequently compared with dNBR-derived burn severity patterns.

5. Results

1. Below is the result of Google Earth Engine script to display DIST-Alerts for two weeks from Sep 1 - 15 2025, along with calculated area and symbology.

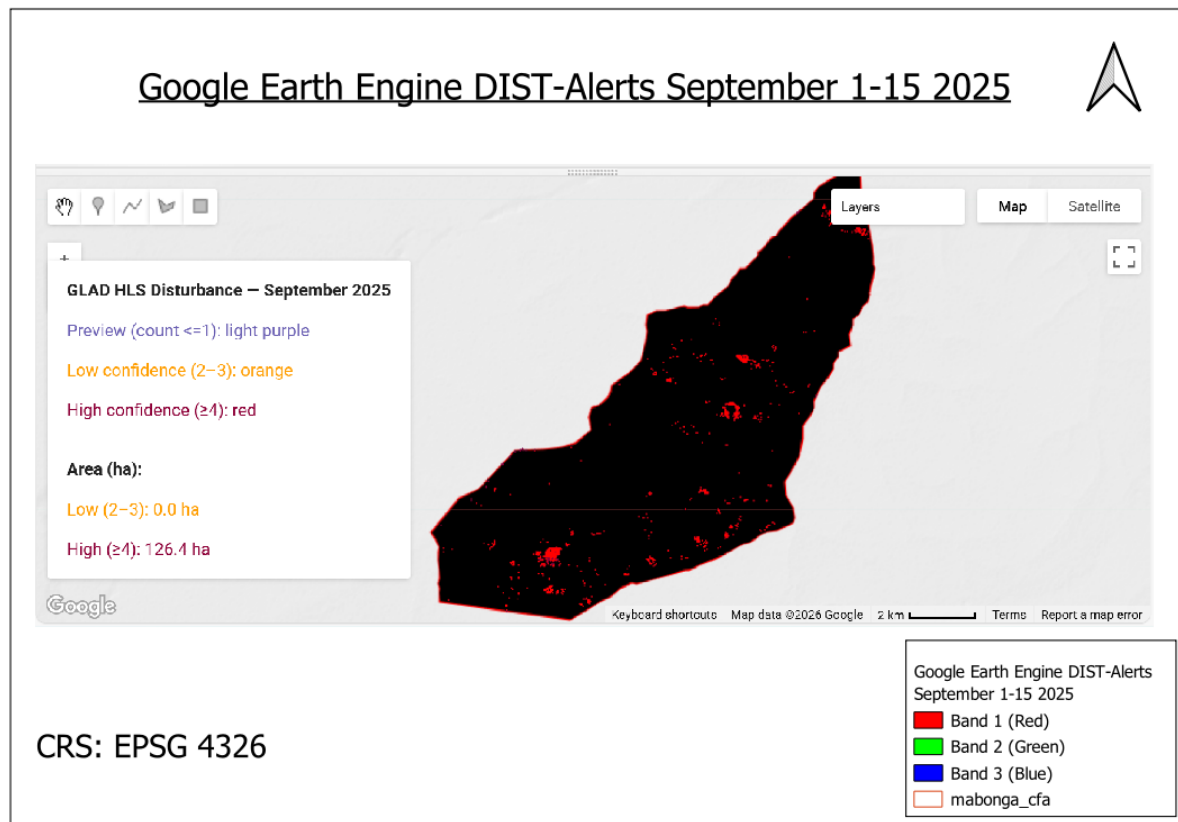


Figure 4 : Disturbance Alerts from Sep1-15,2025 visualized from data in Google Earth Engine

Figure 4 presents the spatial distribution of vegetation disturbance alerts detected between September 1 and 15, 2025. These alerts indicate potential fire-induced disturbance events occurring within the Mabonga Community Forest Area (CFA) during the peak dry season.

2. For the same two-week period, the figure with the dNBR data (Figure 5).

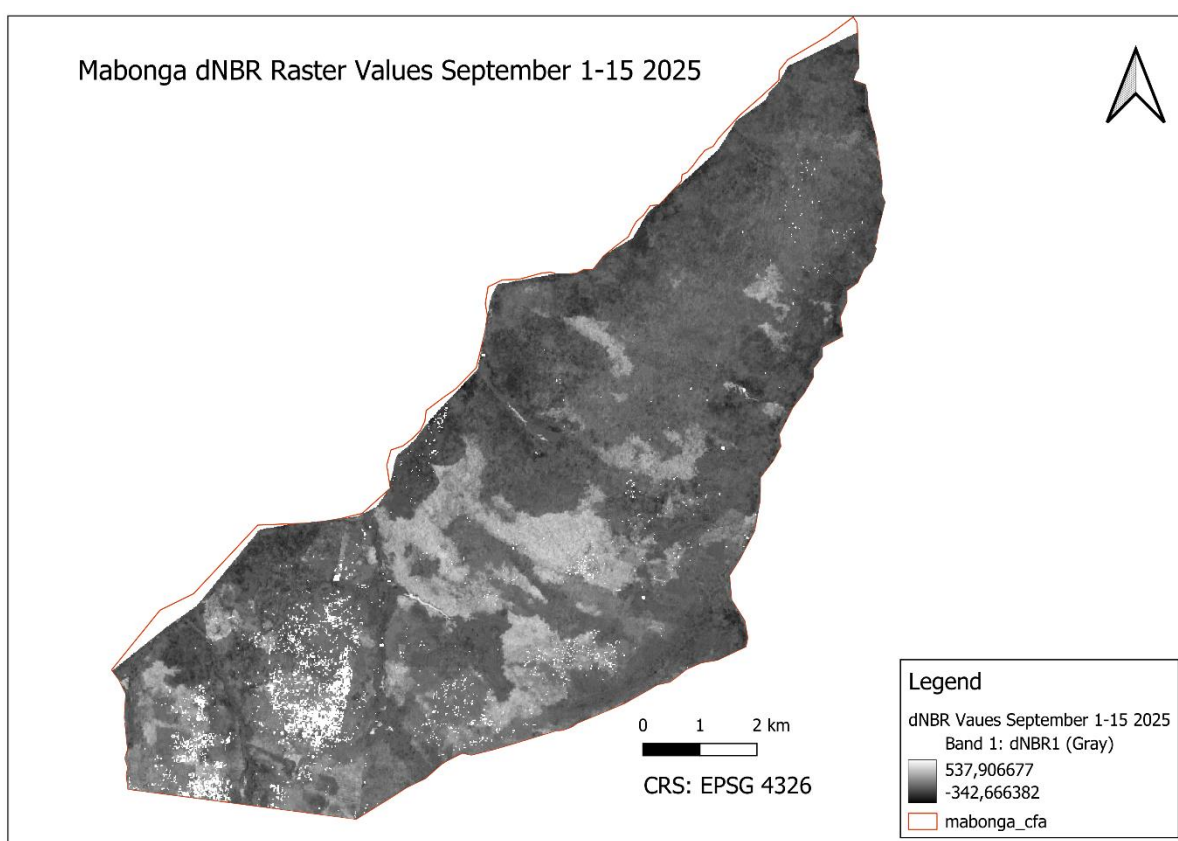


Figure 5 : dNBR data from Sep1-15 2025 visualized in QGIS

Figure 5 illustrates the spatial distribution of burn severity derived from the dNBR index for the same monitoring period. Positive dNBR values indicate burned or disturbed vegetation, while negative values suggest post-disturbance vegetation recovery.

3. Table displaying the dNBR statistics for different time periods for all the DIST-Alerts

DIST-Alert confidence	1,2,3		
dNBR	median	max	min
July (1-15)	64.7698	123.2264	27.5565
July (16-30)	27.2413	164.3185	-17.5714
August (1-15)	67.5632	229.0941	-38.8464
August (16-31)	179.0499	440.7015	-53.7104
Sept (1-15)	38.0014	335.0517	-241.5556
Sept (16-30)	-30.0617	425.3597	-177.7506
Oct (1-15)	-13.9556	345.2379	-372.5342
Oct (16-30)	-131.0139	165.8379	-504.1579

Table 3 : dNBR statistics for DIST-Alert pixels across bi-weekly monitoring periods (July–October 2025).

According to the USGS burn severity classification, positive dNBR values exceeding 0.27 typically correspond to moderate to high severity disturbance, while negative values indicate vegetation recovery.

Interpretation of dNBR Temporal Trends

The median dNBR values increased from July to mid-August, reaching a peak of 179.05 during the August 16–31 monitoring period, indicating the occurrence of high-intensity disturbance events within the CFA. From September onwards, median dNBR values declined sharply and became negative by late September and October. Negative dNBR values typically indicate post-disturbance vegetation regrowth or recovery following fire-induced disturbance. This suggests that previously affected areas began to undergo ecological recovery towards the end of the dry season.

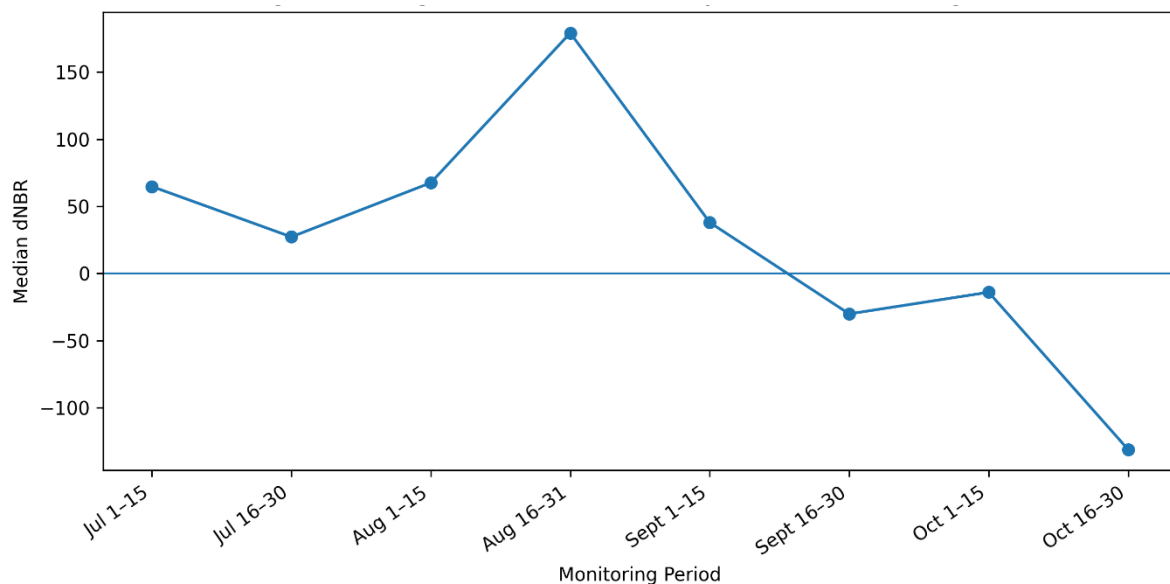


Figure 6: Change in Disturbance Intensity over Time in Mabonga Community Forest Area (CFA)

Figure 6 shows the temporal variation in median dNBR values across bi-weekly monitoring periods from July to October 2025. An increase in median dNBR values between July and mid-August indicates rising disturbance intensity, followed by declining values in September and October, suggesting vegetation recovery after disturbance events.

4. Table with the DIST-Alert area in hectares and the area in percentage (area of the DIST-Alert divided by area of the Mabonga Community Forest Area (CFA) polygon) by period

The total area of the Mabonga Community Forest Area (CFA) Community Forest Area (CFA) is 8,199.71 ha. DIST-Alert area percentages were calculated as the proportion of alert-affected area relative to the

total CFA extent for each monitoring period. Between July and October, a cumulative 462.45 ha (5.64%) of the Mabonga Community Forest Area (CFA) was affected by forest disturbance alerts.

DIST-Alert confidence	1,2,3		
DIST-Alert area	area (ha)	area (%)	Cumulative Area(%)
July (1-15)	0.8204	0.0100	0.0100
July (16-30)	0.9604	0.0117	0.0217
August (1-15)	5.6504	0.0689	0.0906
August (16-31)	64.477	0.7863	0.8769
Sept (1-15)	126.4293	1.5429	2.4198
Sept (16-30)	153.9817	1.8779	4.2977
Oct (1-15)	66.9639	0.8167	5.1144
Oct (16-30)	43.1636	0.5264	5.6408

Table 4 : Calculated Area of DIST-Alerts in hactres and total percentage

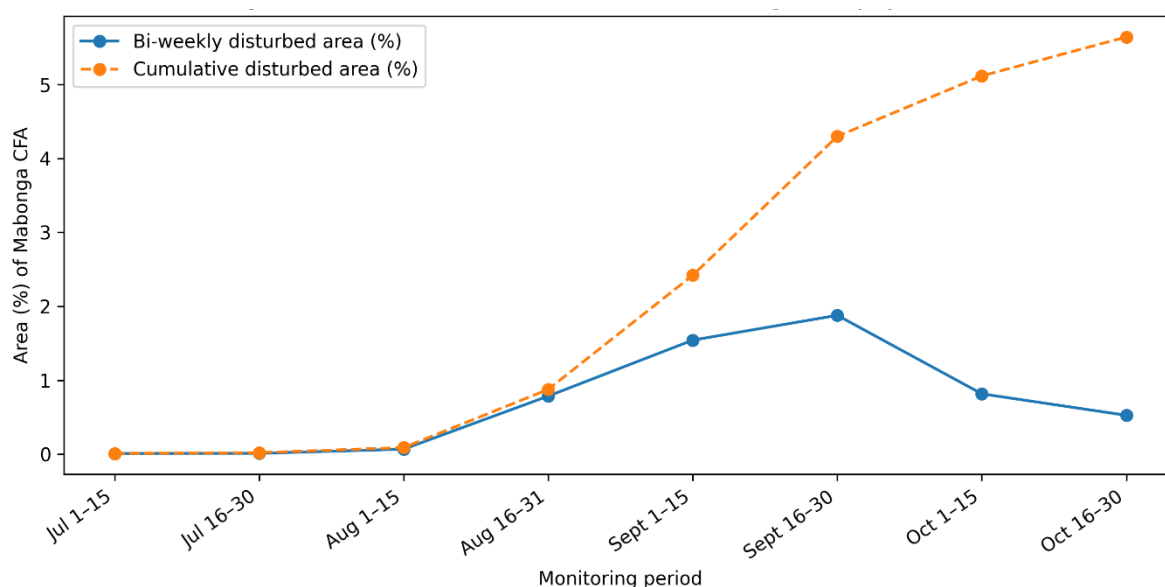


Figure 7: DIST-Alert affected area as % of Mabonga Community Forest Area (CFA), July-Oct 2025

Figure 7 illustrates the temporal variation in DIST-Alert affected area expressed as a percentage of the total Mabonga Community Forest Area (CFA) extent. The increase in disturbed area during August and September reflects heightened fire activity during the dry season.

Temporal Variation in Disturbance Extent

The spatial extent of disturbance alerts increased progressively from July (0.01%) to a peak in mid-September (1.88%). This period coincides with the peak dry season in Muchinga Province, when fire incidence and land-clearing activities are typically highest. Following September, the disturbed area

declined in October, suggesting reduced disturbance activity or stabilization of previously affected areas.

5. Alerts along with their dNBR values visualized in Google Earth pro to visually comprehend the meaning of the DIST-alerts

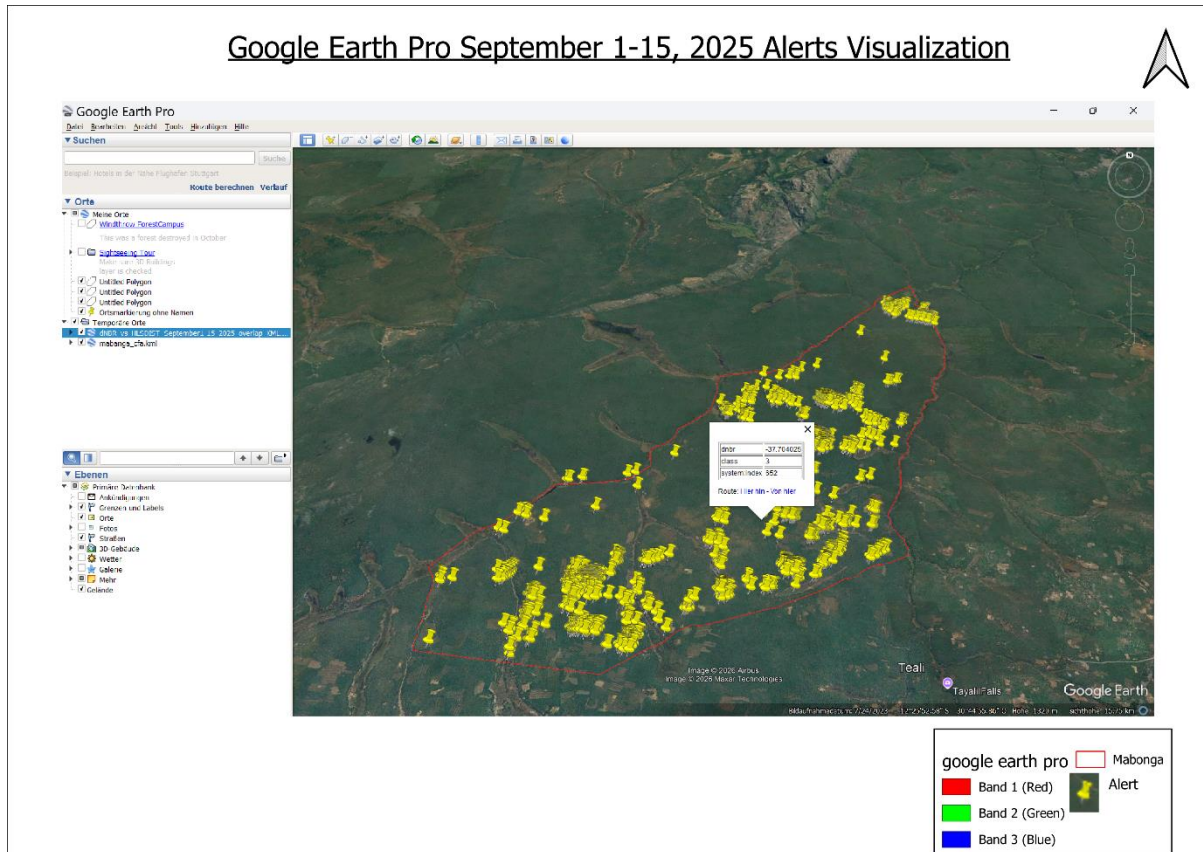


Figure 8 : Visualizing DIST-Alerts along with dNBR values in Google Earth Pro

Figure 8 displays the spatial overlap between DIST-Alert locations and corresponding dNBR values visualized in Google Earth Pro. This visualization was used to assess whether high-confidence alerts corresponded with elevated burn severity values.

6. Discussion

Using the NASA LP DAAC website, initially separate bands for specific dates were downloaded manually to reproduce alerts in the QGIS and though the efforts were rewarding, the time taken to drive alerts manually restricted it to limited analysis and more hard drive space. It was then later decided to analyse the DIST-Alerts with the help of Global Forest Watch, however, because of data unavailability prior to 2025, it had halted the results and hence the best feasible option was generating alerts

through Google Earth Engine script as the specific bands from GLAD Datasets could be downloaded using a script, and analysis could be performed way faster without occupying a lot of computer space and time. Not only that, but from Google Earth Engine, multiple analysis could be performed very efficiently and very timely.

Similarities:

- Where there were high negative dNBR values with high confidence alert values, there was obvious vegetation recovery.

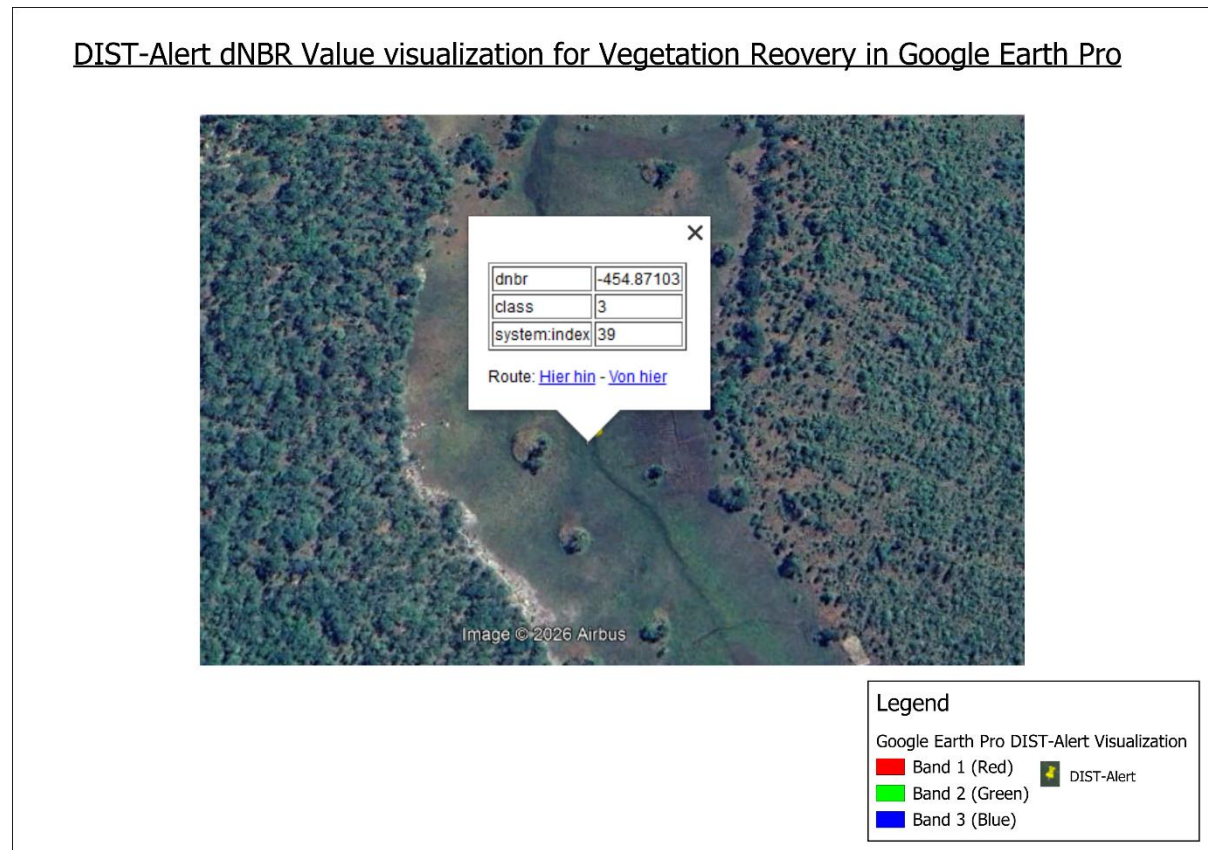


Figure 9: DIST-Alert dNBR Value Visualization for Vegetation Recovery in Google Earth Pro

Figure 9 highlights locations where negative dNBR values coincide with high-confidence DIST-Alerts, indicating post-disturbance vegetation recovery within previously affected forest areas.

- Where there were low negative values were slight usual forest disturbances with also low confidence alert values.

- Where there were high positive values with high confidence alert values, showed forest fire regions and huge burns inside the forests.

DIST-Alert dNBR Value visualization for Forest Burn in Google Earth Pro

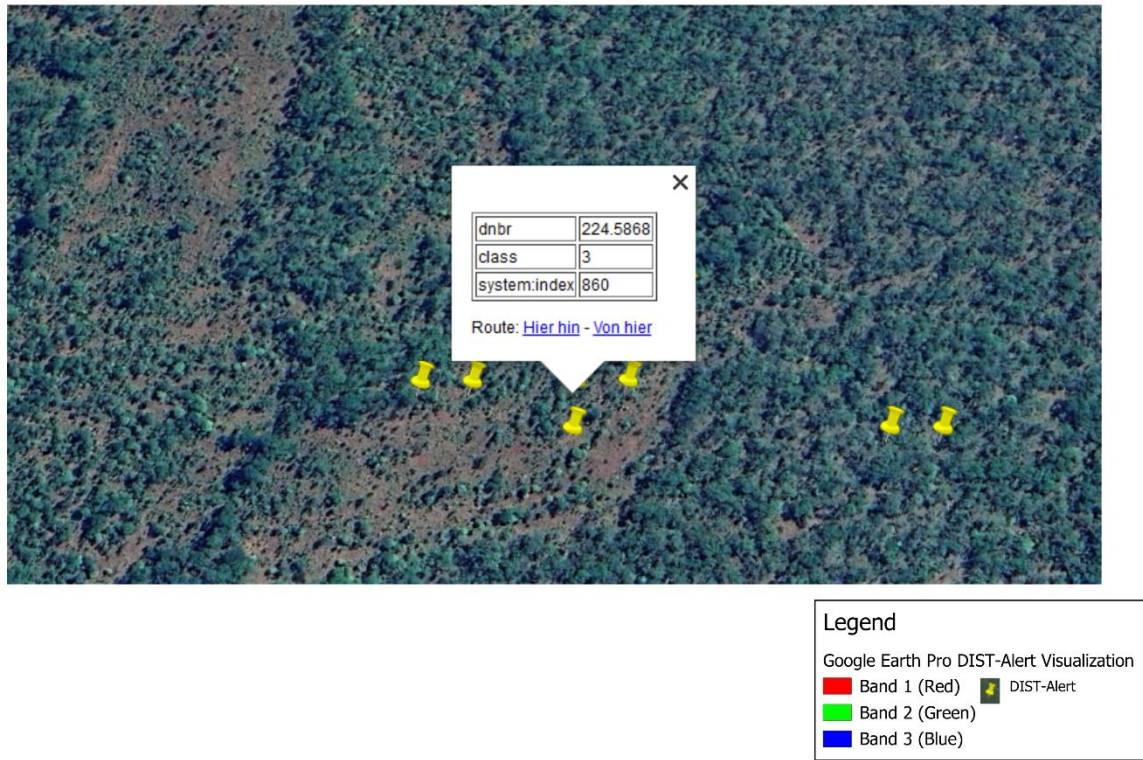


Figure 10: DIST-Alert dNBR Value visualization for Forest Burn in Google Earth Pro

Figure 10 shows areas where high positive dNBR values correspond with high-confidence DIST-Alerts, suggesting the occurrence of fire-induced forest disturbance within the Mabonga Community Forest Area (CFA).

DIST-Alert dNBR Value visualization for Forest Burn in Google Earth Pro

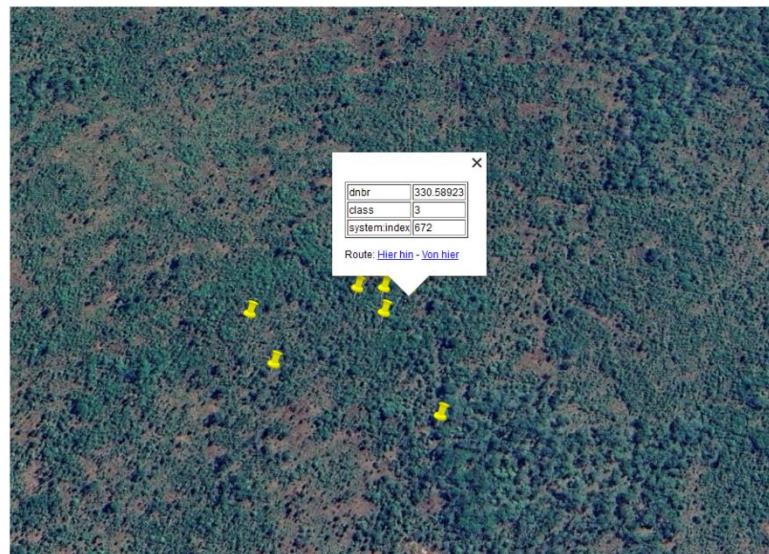


Figure 11 : DIST-Alert dNBR Value Visualization for Forest Burn in google Earth Prot

Figure 11 illustrates regions of severe vegetation disturbance where strong positive dNBR values align with confirmed DIST-Alert detections, indicating high-intensity burn events during the study period.

A general correspondence between high-confidence DIST-Alerts and elevated positive dNBR values was observed across multiple monitoring periods, suggesting that DIST-Alert data provides a reliable near-real-time proxy for detecting fire-induced forest disturbances in Miombo ecosystems.

Differences:

- Where there were low negative or positive dNBR values, the difference was hard to visualize at many places in the Google Earth pro Application.

Reasons for Discrepancies and Patterns:

- Areas with forest fires could be natural forest fires in the region or done by communities for their personal gains.
- Areas with high negative dNBR values could be natural vegetation growth after rain.

DIST-Alert limitations:

- Since the NASA Website is not properly updated, searching area and datasets were a harder part requiring a lot of time and effort for downloading the data and later exploring it.
- DIST-Alerts from Global Forest Watch and Google Earth Engine could not be processed for years prior to 2025 due to limited data availability and hence any kind of analysis could not be conducted before 2025 even though raw data was available at NASA LP DAAC website.
- Where there were low negative or positive dNBR values, the difference was hard to visualize at many places in the Google Earth pro Application.

7. Data Access

The collective data results for DIST-Alerts and dNBR maps used have been shared to be used for future research. Shapefile: Mabonga Community Forest Area (CFA) shapefile has been used to clip the data to the Area of Interest and is also attached in this [link](#) with the report. The dNBR maps used in the research project have been shared for users in the attached [link](#). All the data exports related to this study have been shared in this [link](#). The script written to create all the results is saved in this [link](#). However, to start the analysis from scratch in the QGIS, the raw data can be downloaded from this [link](#).

8. Conclusion

This study demonstrates the applicability of NASA OPERA DIST-Alert data for near-real-time monitoring of fire-induced disturbances within the Mabonga Community Forest Area (CFA). Temporal analysis revealed peak disturbance activity during August–September 2025, followed by vegetation recovery trends in October as indicated by declining dNBR values. The observed spatial agreement between DIST-Alert detections and dNBR-derived burn severity suggests that DIST-Alert can serve as an effective operational tool for fire monitoring and forest management in Miombo woodland ecosystems.

9. References

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